



Project Proposal

BC Personal Health Management Platform

An AI-Powered Healthcare Navigation and Analytics System

CSIS 4495 - 002: Applied Research Project
Winter 2026

Prepared For:

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Date: January 26, 2026

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Contents

1. INTRODUCTION.....	3
1.1 Project Overview.....	3
1.2 Problem Statement & Objectives.....	3
1.2.1 Problem Statement	3
1.2.2 Project Objectives	4
1.2.3 Research Questions.....	4
1.3 Review of Related Literature	4
1.3.1 Research Gaps Addressed	5
1.4 Significance of the Study	6
1.4.1 Healthcare Impact	6
1.4.2 Technical Contributions.....	6
1.4.3 Professional Development.....	6
2. PROPOSED RESEARCH PROJECT	7
2.1 Design and Objectives.....	7
2.1.1 Feature 1: Symptom Assessment Chatbot (PRIMARY FOCUS).....	7
2.1.2 Feature 2: Personal Health Dashboard (PRIMARY FOCUS - ML Analytics).....	7
2.1.3 Feature 3: BC Healthcare Facility Finder	7
2.1.4 Feature 4: Health Education Library	8
2.1.5 Feature 5: System Analytics Dashboard (PRIMARY FOCUS - ML Analytics).....	8
2.2 Methodology & Key Features.....	8
2.2.1 Hybrid Classification Approach	8
2.2.2 OpenAI API Integration	9
2.2.3 ML Analytics Techniques.....	9
2.2.4 Synthetic Data Strategy	9
2.3 Technical Architecture	10
2.3.1 The Hybrid Approach: OpenAI + Rules + ML	10
2.3.2 System Flow Diagram	10
2.3.3 Why This Architecture?.....	11
2.3.4 Component Summary	12
2.3.5 Data Flow for Dashboards.....	12
2.4 Technologies.....	12
2.5 Expected Outcomes.....	14
2.5.1 Primary Deliverables.....	14
2.5.2 Success Metrics.....	14
3. PROJECT PLANNING & TIMELINE.....	15
4. PROJECT CONTRACT.....	17

5. AI USE SECTION	18
5.1 AI Tools Usage Summary.....	18
5.2 Table of AI Tools and Specific Use	18
5.3 Statement of Originality	19
5.4 Prompt History	19
6. WORK DATE/HOURS LOG.....	20
7. GITHUB REPOSITORY.....	22
8. REFERENCES.....	23
APPENDIX: AI PROMPT HISTORY.....	24
Prompt Session 1: Clarifying Hybrid Approach	24
Prompt Session 2: Understanding OpenAI API Options	24
Prompt Session 3: ML Analytics Techniques.....	25
Prompt Session 4: Data Source Question.....	25
Prompt Session 5: Synthetic Data for Demo.....	26
Prompt Session 6: Document Formatting	26
Prompt Session 7: System Flow Diagram Generation	27
Prompt Session 8: Architecture Overview Diagram Generation.....	27
Summary of AI Usage	28

1. INTRODUCTION

1.1 Project Overview

Across British Columbia, emergency departments are under increasing strain as more patients seek care while resources remain limited. The Fraser Health Authority caters to a population of over 1.9 million, and emergency departments there frequently run into very long waiting times. During my co-op stint at Richmond Hospital Emergency Department, I witnessed that about 15-20% of those coming to the emergency were not clear if their symptoms needed emergency care, urgent care, or if they could have been taken care of at home, thus resulting in inefficient use of resources and unnecessary visits to the ED.

The BC Personal Health Management Platform bridges this disconnect by offering a complete, multi-featured web application that combines symptom checking, personalized health tracking, and population, level analytics. This platform goes beyond merely symptom checkers, offering a comprehensive solution by integrating five interconnected features:

- ***Symptom Assessment Chatbot*** with OpenAI integration and ML classification
- ***Personal Health Dashboard*** with individual-level ML analytics
- ***BC Healthcare Facility Finder*** with interactive mapping
- ***Health Education Library*** with evidence-based guidance
- ***System Analytics Dashboard*** with population-level ML insights

The platform employs a hybrid architecture strategy that integrates Python/Streamlit for the main application features and Power BI for highly professional data visualization. This technique enables me to showcase my skills in both programming and business intelligence tools which fit perfectly with my Data Analytics specialization as well as my career goals in healthcare analytics.

With the help of Professor Kandhari's advice (Meeting #2, January 18, 2026), the project focuses on three main priorities: (1) complete workflow functionality, (2) OpenAI integration for natural language processing, and (3) machine learning analytics. The chatbot is one of the major features in the entire health management ecosystem, making sure that the project really brings out data analytics capabilities, rather than just being a software development project.

1.2 Problem Statement & Objectives

1.2.1 Problem Statement

BC's healthcare system faces three interconnected challenges:

1. **Emergency Department Overcrowding:** Research shows that 15 - 30% of visit to the ED can be redirected to other settings. However, patients are not equipped with accessible tools to decide on the level of care that is appropriate for them.

2. Limited Decision Support: When the patients lack support, they either trust the online sources which may not be reliable or visit the ED as a default. Currently, there is no mobile-accessible, evidence-based guide specific to BC resources that patients can refer to.

3. Lack of Personal Health Insights: People cannot easily monitor their health conditions over a period of time, and they do not know when their frequently occurring symptoms are serious enough to warrant a doctor's visit.

1.2.2 Project Objectives

Primary Objectives (Core Deliverables):

- Develop *symptom assessment chatbot* capable of OpenAI API integration for natural language understanding
- Implement *hybrid classification approach*: rule-based logic for critical symptoms (based on CTAS guidelines) + ML model for ambiguous cases
- Set up *Power BI dashboards* for personal health tracking (Feature 2) and system analytics (Feature 5) with ML-driven insights
- Craft a *BC healthcare facility locator* providing interactive map visualization
- Host complete *multi-page web application* accessible via public URL
- Ensure classification accuracy of $\geq 75\%$ together with sensitivity of $\geq 90\%$ for the emergency cases

Secondary Objectives (Stretch Goals):

- Implement UI for conversational agent to ask questions about symptoms for further clarification
- Implement advanced ML analytics: clustering, anomaly detection, forecasting
- Conduct user testing with classmates for feedback

1.2.3 Research Questions

1. Can a hybrid approach (rule-based + ML) accurately classify symptom urgency and, at the same time, ensure the critical cases are already treated as safety cases?
2. What is the contribution of OpenAI API integration in enhancing natural language understanding of unstructured symptom descriptions?
3. What ML analytics techniques (clustering, trend detection, anomaly detection) can deliver significant health insights at the levels of individual and population?
4. How can the principles of explainable AI be used to establish confidence in automated triage suggestions?

1.3 Review of Related Literature

Transformer-based model Medical NLP has progressed rapidly. Medical entity recognition by BioBERT was shown by Lee et al. (2020) to attain F1 scores over 0.90, indicating that domain-specific language

models perform better than general NLP ones in healthcare contexts. The advent of Large Language Models such as GPT-4 has made a major impact on medical text processing, allowing highly advanced natural language understanding without requiring much domain-specific training.

Symptom checkers are one of the types of systems that can be helpful in symptom assessment. ML-based symptom checkers can determine triage decisions with a 75-85% accuracy rate as compared to emergency physicians for common patient cases, as demonstrated by Ceney et al. (2021). Besides that, Semigran et al. (2015) noted a significant variation in the results of 23 tested symptom checker software from 34-80% accuracy, thus indicating irregular quality and a necessity for hybrid approaches that combine ML's flexibility with rule, based safety mechanisms. Moreover, machine learning has also demonstrated the capability of predicting hospital admission decisions at emergency department triage, hence illustrating the potential of ML-based clinical decision support tools (Hong et al., 2018).

The Canadian Triage and Acuity Scale (CTAS), a five-level standardized system used throughout Canadian hospitals, was developed by the Canadian Emergency Department Triage & Acuity Scale Implementation Guidelines (2008). Though skilled clinicians are needed to perform the CTAS, its well-organized decision logic offers a basis for computational triage methods, especially for critical symptoms to be managed using rules only.

Population-level health analytics has become an important tool for public health surveillance. Patient subgroups with similar symptoms have been identified through the use of clustering methods (such as K-means and hierarchical clustering) (Liao et al., 2016). Besides that, deep learning techniques have been used on EHRs to identify patients' future health trajectories, thus proving the efficiency of ML healthcare analytics (Miotto et al., 2016). Also, time series analysis and anomaly detection have been used not only for identifying disease outbreaks but also planning for healthcare resources.

Explainability remains a key feature of healthcare AI. Holzinger et al. (2017) point out that it is not enough to rely on "black box" ML models for clinical support; therefore, clinicians should use interpretable models or post-hoc explanation tools such as SHAP and LIME. This work takes into account the healthcare explainability issue by delivering clear rule-based reasoning for urgent scenarios together with the feature importance explanations for ML-derived predictions.

1.3.1 Research Gaps Addressed

By analyzing the literature, it was possible to identify several limitations in the current symptom assessment tools that this project wants to address.

- Most symptom checkers use either rules OR ML—this project integrates both to provide safety and flexibility
- Limited integration of OpenAI/LLM capabilities in patient-facing triage tools
- Few implementations include population-level ML analytics alongside individual assessment
- Lack of BC-specific healthcare contextualization in existing tools

1.4 Significance of the Study

1.4.1 Healthcare Impact

Through the lens of healthcare, my project tackles the real issues that I witnessed first-hand during my co-op at Richmond Hospital where about 15-20% of patients coming to the ED wouldn't know the right level of care to go for:

- Emergency Department Efficiency: Providing evidence, based guidance will most likely lead to a drop in the number of people going to the ED when it is not necessary
- Patient Empowerment: An easy-to-use tool for making well, informed decisions on when to seek care by also tracking personal health changes
- Population Health Insights: Aggregated analytics can identify symptom trends and potential outbreaks
- BC Context: Clear integration with HealthLinkBC, Fraser Health urgent care, 811/911 protocols

1.4.2 Technical Contributions

Technically this work shows a healthcare solution enabled by the integration of several AI/ML technologies:

- Hybrid triage architecture combining rule-based safety with ML flexibility
- OpenAI API integration for natural language symptom processing
- ML analytics pipeline for both individual and population-level insights
- Integration of Streamlit application with embedded Power BI dashboards

1.4.3 Professional Development

Reflecting a career standpoint, this project is a perfect match with BC healthcare data science job requirements and it is a great end-to-end ML system development demonstration to have in my portfolio:

- Addresses BC healthcare data science job requirements: NLP, ML, clinical analytics
- Demonstrates proficiency in Python, Power BI, and cloud deployment
- Creates portfolio showcasing end-to-end ML system development
- Directly relevant to Fraser Health Data Scientist and similar roles

2. PROPOSED RESEARCH PROJECT

2.1 Design and Objectives

The Personal Health Management Platform of BC is designed around a modular architecture with five interoperable features. AI/ML components form the backbone of the platform as directed by Professor Priya:

2.1.1 Feature 1: Symptom Assessment Chatbot (PRIMARY FOCUS)

The chatbot integrates OpenAI with custom ML classification in a hybrid way:

- **Layer 1 (Rule-Based):** Critical symptoms have been pre-set in accordance with CTAS guidelines. To ensure safety, emergency recommendations are automatically given for symptoms like chest pain and shortness of breath, without depending on ML predictions.
- **Layer 2 (ML Classification):** Using a trained Random Forest classifier, non-critical symptoms are predicted for their level of urgency (Emergency, Urgent, Routine, Self, Care), based on symptom patterns from medical datasets on Kaggle.
- **OpenAI Integration:** Natural language input from the user is transformed into structured symptom information (symptoms, duration, severity) through the OpenAI API before classification.
- **Explainability:** Each recommendation comes with a clear explanation that indicates the symptoms which justified the classification.

2.1.2 Feature 2: Personal Health Dashboard (PRIMARY FOCUS - ML Analytics)

A Power BI dashboard displaying the assessment history of the individual user and ML-driven insights, including:

- **Trend Detection:** Using time series analysis to uncover patterns in the symptom history of a user
- **Association Rules:** Recognizing symptom co-occurrences (e.g., "When you report headache, you also report fatigue 75% of the time")
- **User Profiling:** Using clustering to relate user's symptom patterns with health profiles
- **Personalized Recommendations:** Derived from the user's past behavior

2.1.3 Feature 3: BC Healthcare Facility Finder

An interactive map pinpointing healthcare facilities in BC:

- Information has been gathered manually from HealthLinkBC and Fraser Health websites
- Displays EDs, urgent cares, walk, in clinics along with their hours, phone numbers, and addresses
- Integration with chatbot recommendations (e.g., "Nearest urgent care: Surrey Memorial")

2.1.4 Feature 4: Health Education Library

A database that can be searched for health topics based on scientific evidence:

- Focus on common health problems and symptoms along with the corresponding care instructions
- Material is obtained from HealthLinkBC, rewritten for clarity, and properly referenced
- Search functionality for quick access to retrieve relevant information

2.1.5 Feature 5: System Analytics Dashboard (PRIMARY FOCUS - ML Analytics)

A Power BI dashboard showing aggregate trends across all users with ML-driven insights:

- **Population Trends:** Time-series demonstrating how frequently symptoms are exhibited
- **K-Means Clustering:** Identifying and differentiating distinct user groups based on symptom patterns
- **Anomaly Detection:** Flagging unusual surges in particular symptoms or geographical areas
- **Forecasting:** Predicting the number of symptoms in the following week
- **Geographic Analysis:** Displaying regional symptom distribution by the heatmap

2.2 Methodology & Key Features

2.2.1 Hybrid Classification Approach

The symptom classification uses a two-layer hybrid approach to ensure both safety and flexibility:

Layer 1: Rule-Based Critical Symptom Detection

Based on CTAS (Canadian Triage and Acuity Scale) guidelines, critical symptom combinations trigger immediate recommendations without ML:

- Chest pain + shortness of breath → CALL 911 IMMEDIATELY
- Face drooping + arm weakness + speech difficulty → CALL 911 (stroke signs)
- Severe bleeding or unconsciousness → CALL 911 IMMEDIATELY
- Fever + stiff neck + light sensitivity → GO TO ER (possible meningitis)

Layer 2: ML Classification for Ambiguous Cases

When no critical rules match, symptoms pass to the ML classifier:

- **Training Data:** Kaggle symptom-disease datasets with manually mapped urgency levels
- **Algorithm:** Random Forest Classifier (primary), with Gradient Boosting as alternative
- **Features:** Symptom presence, severity scores, duration, symptom combinations
- **Output:** Four-class prediction (Emergency, Urgent, Routine, Self-Care) with confidence scores

2.2.2 OpenAI API Integration

OpenAI's GPT model processes unstructured user input to extract structured symptom information (OpenAI, 2024):

- User types: "I've had chest pain for 2 hours and feel dizzy"
- OpenAI extracts: {symptoms: ["chest pain", "dizziness"], duration: "2 hours"}
- Structured data passes to Layer 1 (rule check) then Layer 2 (ML) if needed

This approach handles natural language variations ("my chest hurts" vs "chest pain" vs "pain in chest") without requiring extensive manual keyword lists.

2.2.3 ML Analytics Techniques

The following ML techniques power Features 2 and 5:

Technique	Feature	Purpose	Library
<i>Time Series Analysis</i>	2, 5	Detect trends in symptom frequency	pandas, Power BI DAX
<i>Association Rules (Apriori)</i>	2, 5	Find symptom co-occurrence patterns	mlxtend
<i>K-Means Clustering</i>	5	Group users by symptom profiles	scikit-learn
<i>Isolation Forest</i>	5	Detect anomalous symptom spikes	scikit-learn
<i>Linear Regression</i>	5	Forecast future symptom volumes	scikit-learn

2.2.4 Synthetic Data Strategy

To enable meaningful dashboard demonstrations, the application will be pre-loaded with synthetic assessment data:

- 5,000 synthetic records simulating 3 months of user activity
- Realistic symptom patterns with embedded trends (e.g., simulated flu outbreak)
- Enables ML analytics to display meaningful insights during demonstrations
- Real user data accumulates on top of synthetic baseline

This approach is standard practice for proof-of-concept applications and will be clearly documented in the final report.

2.3 Technical Architecture

The platform implements a hybrid architecture that sequentially integrates three AI/ML components. The decision to have OpenAI integration as a major technical element was based on the instructor's direction (Meeting #2).

2.3.1 The Hybrid Approach: OpenAI + Rules + ML

A major signature of this project is the three, layer hybrid architecture for symptom classification. Conventional symptom checkers generally rely on ONLY rules or ONLY ML, whereas this system is using the following combinations:

- OpenAI API: Deciphers natural language (handles variations like "my chest hurts" vs "chest pain")
- Rule-Based Logic: Detects critical emergencies with 100% reliability (never misses life-threatening symptoms)
- ML Classification: Predicts urgency for ambiguous cases where trained patterns are used as a basis

2.3.2 System Flow Diagram

Figure 2.3.2 gives a visual representation of the three, layer hybrid classification system. Suppose a user provides symptoms in natural language, then **Step 1 (OpenAI API)** breaks down the input to get the structured data containing symptoms, duration, and severity. The structured information is then sent to **Step 2 (Rule-Based Safety Check)**, which uses CTAS-based rules to identify symptom patterns that indicate critical issues like chest pain with difficulty in breathing or stroke symptoms. If such a critical match is found, the system sends back an **Emergency** response right away instructing the user to call 911. In the absence of such a critical match, the input goes to **Step 3 (ML Model Classification)** where a trained Random Forest classifier identifies one of **four levels of urgency: Emergency, Urgent, Routine, or Self-Care**. The last result gives the user the urgency level marked with a color, a simple explanation of why this advice was given, and healthcare resources in BC such as HealthLinkBC, 811, or a nearby facility.

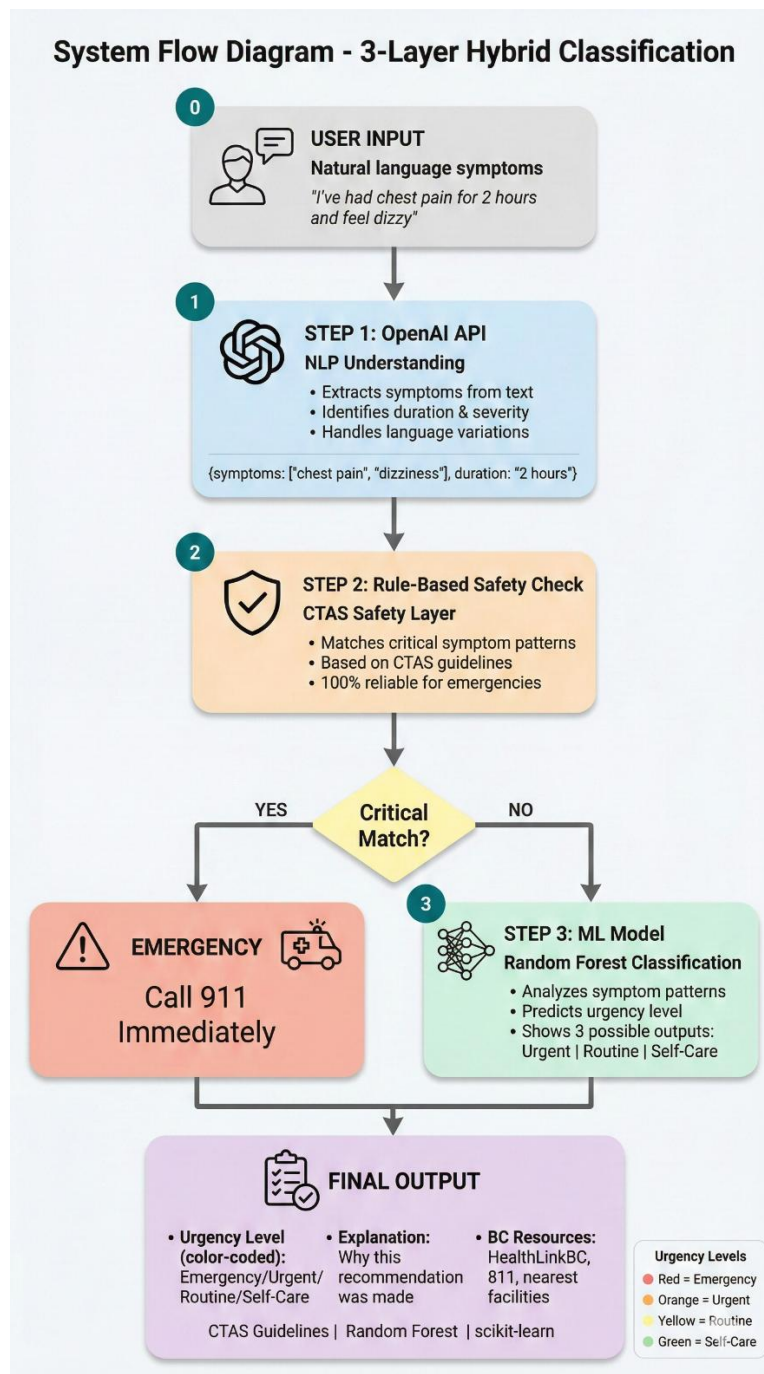


Fig 2.3.2 – System Flow Diagram (Generated using Gemini image generator)

2.3.3 Why This Architecture?

This three-layer approach provides a solution to the weaknesses in the simpler approaches:

Approach	Limitation	How This Project Solves It
Traditional NLP Only	Must manually code EVERY way users describe symptoms	OpenAI understands all variations automatically

	("chest pain" vs "my heart hurts" vs "tightness in chest")	
ML Only	May misclassify critical emergencies; "black box" predictions	Rule-based layer ensures critical symptoms ALWAYS trigger emergency response
Rules Only	Cannot handle ambiguous or complex symptom combinations	ML model learns patterns from training data for non-critical cases

2.3.4 Component Summary

Component	Technology	Purpose
Step 1: NLP Understanding	OpenAI API (GPT-3.5-turbo)	Understand natural language, extract structured symptom data
Step 2: Safety Check	Rule-based logic (Python)	Catch critical emergencies based on CTAS guidelines
Step 3: Classification	Random Forest (scikit-learn)	Predict urgency level for non-critical symptoms

2.3.5 Data Flow for Dashboards

Besides how the chatbot works, assessment data flows to the analytics components:

1. A record of each assessment is kept in an SQLite database and then exported as a CSV file
2. The CSV files are used as a data source for the Power BI dashboards
3. Python ML scripts further process the data for clustering, anomaly detection, and forecasting.
4. The output is used for the visuals in Power BI (Features 2 and 5).

2.4 Technologies

Figure 2.4 presents the overall architecture and data flow of the BC Personal Health Management Platform. The system follows a layered design where users interact with the **Streamlit Cloud** application through the internet. The application layer contains all five platform features—Symptom Chatbot, Personal Dashboard, Facility Finder, Education Library, and System Analytics—supported by four core processing components: **Python 3.10+** being the programming base, **NLTK and spaCy** for NLP preprocessing, **scikit-learn** as the ML engine, and **Rule-Based Logic** for safety-critical decisions. Two external services are used to enhance platform functionalities: **OpenAI API** enables natural language understanding to extract symptoms, and **Power BI Service** offers user-friendly embedded dashboards.

The data layer is where **SQLite** locally saves the assessment records that are subsequently transformed into **CSV files** for Power BI integration, while **GitHub** is used for version control.

The BC Personal Health Management Platform draws from three comprehensive data sources: **Kaggle** symptom-disease datasets for ML model training, **HealthLinkBC** for local BC-specific health guidance, and **CTAS** (Canadian Triage and Acuity Scale) for clinically validated triage rules.

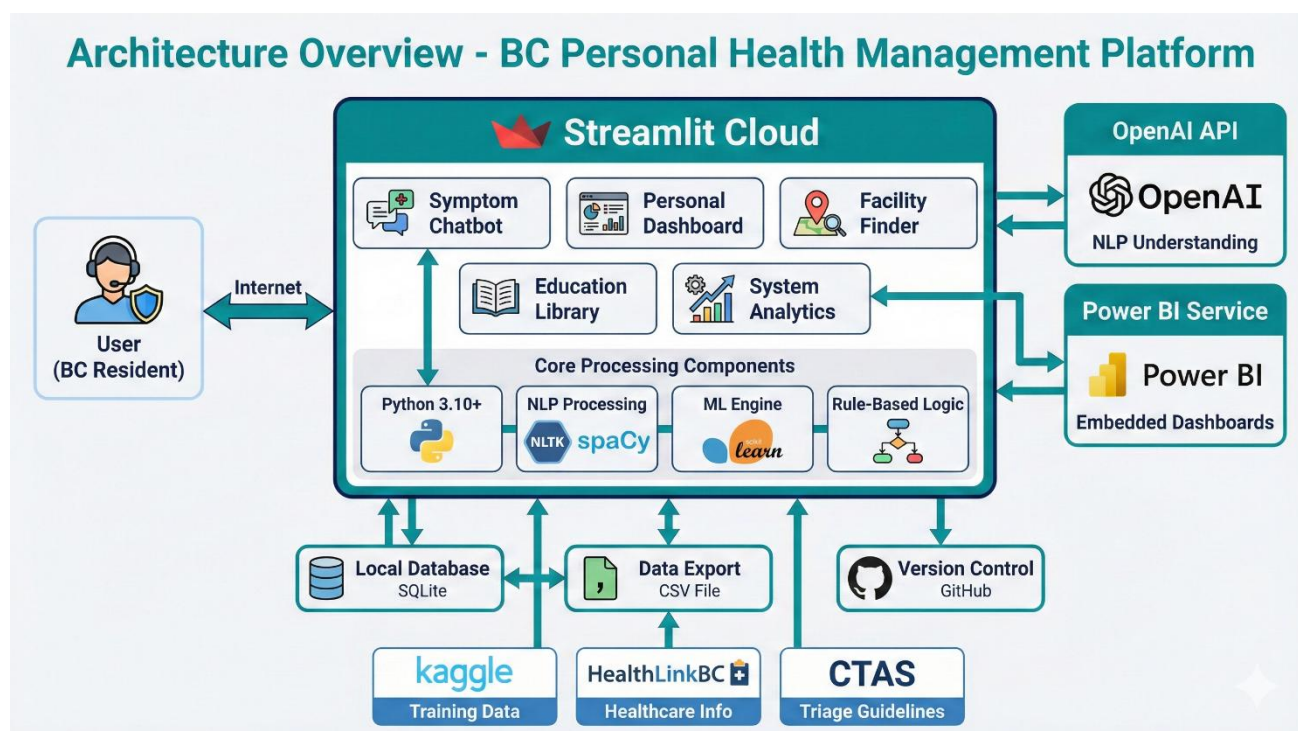


Fig 2.4 – Architecture Overview showing technology stack and data flow (Generated Using Gemini image generator)

The following table provides additional details on each technology component:

Category	Technology	Purpose
Operating System	Windows 11 (Dev), Linux Ubuntu (Deploy)	Development and deployment environments
Programming Language	Python 3.10+	Core application, ML, NLP processing
Web Framework	Streamlit 1.30+	Multi-page web application
AI/NLP	OpenAI API (GPT-3.5-turbo)	Natural language symptom extraction
NLP Libraries	NLTK, spaCy	Text preprocessing, tokenization
Machine Learning	scikit-learn, mlxtend	Classification, clustering, association rules
Data Visualization	Power BI Desktop + Service	Embedded interactive dashboards
Database	SQLite + CSV exports	Data storage and Power BI integration

Deployment	Streamlit Cloud + Power BI Service	Public URL access
Version Control	GitHub	Code repository and collaboration

2.5 Expected Outcomes

2.5.1 Primary Deliverables

1. **Deployed Multi-Page Web Application:** Public URL with 5 integrated features
2. **Hybrid Classification System:** Rule-based + ML achieving $\geq 75\%$ accuracy, $\geq 90\%$ emergency sensitivity
3. **OpenAI Integration:** Natural language processing for symptom extraction
4. **Power BI Dashboards:** Personal (Feature 2) and System (Feature 5) analytics with ML insights
5. **Complete Documentation:** Proposal, Midterm Report, Final Report, Demo Videos

2.5.2 Success Metrics

Metric	Target	Measurement Method
Classification Accuracy	$\geq 75\%$ overall	Confusion matrix on test set
Emergency Sensitivity	$\geq 90\%$	Recall for emergency class
Response Time	<3 seconds	End-to-end processing time
Application Uptime	99%+	Streamlit Cloud monitoring
Dashboard Functionality	All visualizations working	Manual testing checklist

3. PROJECT PLANNING & TIMELINE

Project Timeline (Gantt Chart)

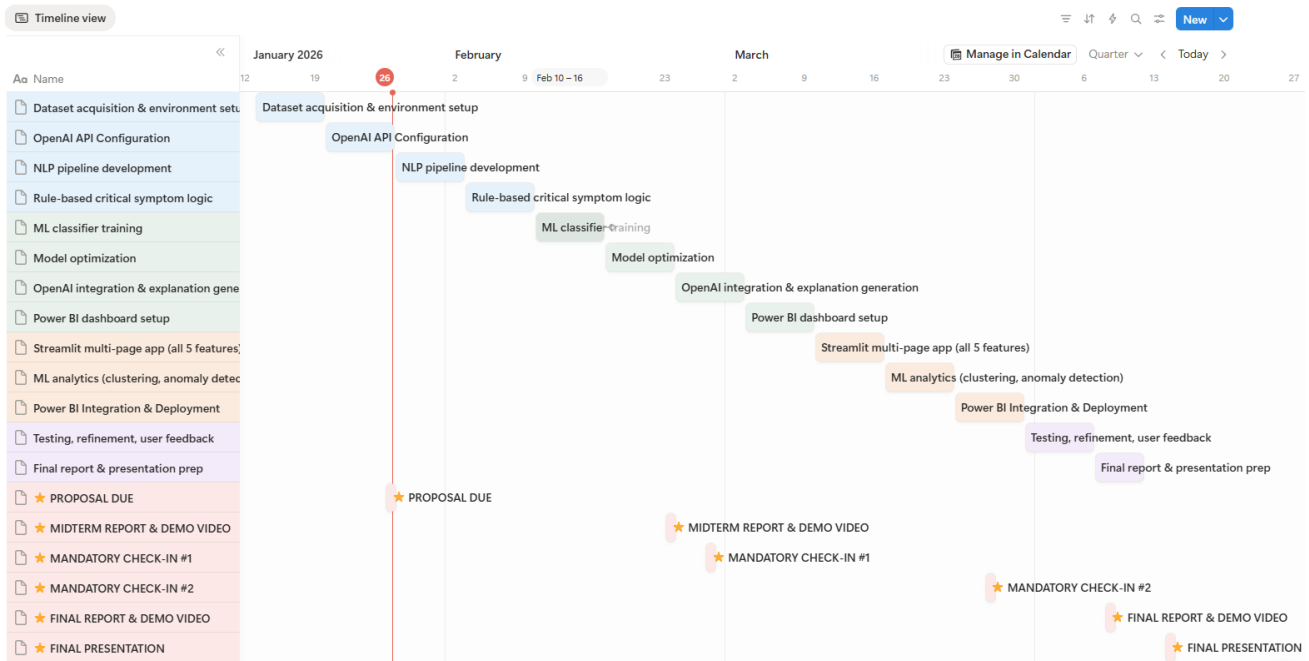


Fig 3 – Project Gantt Chart (Generated using project management software - Notion)

Milestone	Due Date	Phase	Deliverables	Tasks to Complete
Proposal Submission	Jan 26, 2026	Phase 1 Jan 13 - Feb 9, 2026	- Project proposal document - Literature review - System architecture diagrams - Project timeline (Gantt chart) - Risk assessment	- Dataset acquisition & environment setup - OpenAI API configuration - NLP pipeline development - Rule-based critical symptom logic
Midterm Report & Demo Video	Feb 23, 2026	Phase 2 Feb 10 - Mar 9, 2026	- Midterm progress report 5-minute demo video - Working NLP pipeline - Rule-based critical symptom logic - Initial ML classifier prototype	- ML classifier training - Model optimization
Mandatory Check-in #1	Feb 27, 2026	Phase 2	Progress presentation to instructor	OpenAI integration & explanation generation

Mandatory Check-in #2	Mar 27, 2026	Feb 10 - Mar 9, 2026	• Live demonstration of hybrid classification • Q&A session on implementation challenges	• Power BI dashboard setup
		Phase 3 Mar 10 - Mar 30, 2026	• Progress presentation to instructor • Live demonstration of Streamlit app • All 5 features functional • Power BI dashboard preview	• Streamlit multi-page app • ML analytics - clustering, anomaly detection • Power BI integration & deployment
Final Report & Demo Video	Apr 8, 2026	Phase 4 Mar 31 - Apr 14, 2026	• Complete final report (all sections) • 10-minute final demo video • Fully deployed Streamlit application • Source code repository (GitHub) • User documentation	• Testing, refinement, user feedback • Final report & presentation prep
Final Presentation	Apr 14, 2026	Phase 4 Mar 31 - Apr 14, 2026	• 15-minute live presentation • Live system demonstration • Technical Q&A readiness • Project reflection & lessons learned	• Final report & presentation prep • All 13 tasks complete

4. PROJECT CONTRACT



January 26, 2026

Padmapriya Arasanipalai Kandhadai

Professor/Advisor

Douglas College

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New Westminster, BC V3M 5Z5

Dear Prof. Priya,

This letter serves as my agreement to develop the **BC Personal Health Management Platform** for my CSIS 4495 Applied Research Project. The platform is a multi-feature web application combining Python/Streamlit with Power BI for healthcare navigation and analytics. The five key features include: (1) a *Symptom Assessment Chatbot* with OpenAI integration and hybrid ML classification, (2) a *Personal Health Dashboard* with ML analytics, (3) a *BC Healthcare Facility Finder*, (4) a *Health Education Library*, and (5) a *System Analytics Dashboard* with population-level ML insights. The system implements a hybrid classification approach using rule-based logic for critical symptoms (based on CTAS guidelines) and machine learning for ambiguous cases, targeting $\geq 75\%$ accuracy with $\geq 90\%$ sensitivity for emergency cases.

I will follow the project timeline in the proposal, delivering: (1) a proof-of-concept with rule-based layer working, (2) a functional chatbot with hybrid classification, (3) a deployed multi-page application with embedded Power BI dashboards, and (4) complete documentation including the final report and presentation by the stated milestones. I am responsible for development, testing, and deployment within the agreed scope, technical requirements, and deadlines. A minimum of eleven (11) hours per week will be dedicated to the project, totaling 144 hours over 13 weeks.

Mandatory check-ins will be attended on February 27, 2026 and March 27, 2026 to track progress and surface any risks. Any scope or schedule changes will be communicated promptly and documented. This agreement reflects my commitment to delivering a functional, well-documented healthcare navigation solution that demonstrates proficiency in OpenAI integration, machine learning classification, and data analytics—aligned with my Data Analytics specialization at Douglas College.

Sincerely Yours,

Desmond Chua

Desmond Chua (300369803)

5. AI USE SECTION

5.1 AI Tools Usage Summary

AI tools were used sparingly during the proposal development phase. Most of the time, I used them to understand better or explain technical concepts and to check if I understood the methods I had already researched correctly. I came up with the main ideas of the project, made the decisions about the architecture, and chose the method based on my experience at Richmond Hospital and my studies in data analytics and machine learning.

5.2 Table of AI Tools and Specific Use

AI Tool	Version/Account	Specific Use	Value I Added
Claude (Anthropic)	Claude.ai, Free account	Clarifying hybrid ML architecture concept I was considering	Validated my approach; refined rule-based vs ML split based on CTAS research I conducted independently
ChatGPT	ChatGPT, Free account	Explaining OpenAI API integration options	Decided on implementation approach; wrote all code logic myself based on OpenAI documentation
Claude (Anthropic)	Claude.ai, Free account	Reviewing ML analytics techniques for dashboards	Selected specific techniques (clustering, anomaly detection) based on my Power BI experience at VCH
Claude (Anthropic)	Claude.ai, Free account	Document formatting assistance for proposal structure	All content written by me; used for formatting tables and structure only
Google Gemini	Gemini, Free account	Generating professional architecture diagrams (Fig 2.3.2 System Flow Diagram and Fig 2.4 Architecture Overview)	Created all diagram content, specifications, and layout requirements myself; provided detailed prompts specifying exact components, data flow, and technical elements based on my system design

5.3 Statement of Originality

Below is a list of original and true components of my work:

- **Project concept:** This idea was derived from a co-op at Richmond Hospital where it was noted that 15-20% of Emergency Department patients were unsure of the level of care they were to receive.
- **Problem identification:** From direct experience in healthcare setting
- **Technical architecture:** Designed based on CSIS 3400 (NLP) coursework and VCH Power BI experience
- **Hybrid approach decision:** Researched CTAS guidelines independently; determined rule-based safety layer was necessary
- **Features selection:** Based on instructor feedback (Meeting #1 and #2) to broaden scope which expanded from symptom assessment chatbot to more features
- **Data source strategy:** Researched Kaggle datasets alongside with CTAS documentation

AI tools were only 'looked up' to confirm my understanding and to figure out the 'how-to' part of the implementation—this is no different from referring to a textbook or shooting a question to a classmate. All code, architectural decisions, and writing are my own work and represent my understanding of the matter.

5.4 Prompt History

The full prompt history detailing all the AI interactions is in the Appendix section. Each entry provides: (1) my initial prompt/question, (2) what the AI response was, and (3) the value I contributed beyond the AI's output. The provided documentation helps to verify the claim that AI-enabled tools have been used only for clarifications and not for content generation purposes.

6. WORK DATE/HOURS LOG

Student: Desmond Chua | Student ID: 300369803

Note: Log updated with granular entries. Broad entries avoided.

Date	Hours	Description of Work Done
Jan 9, 2026	2.0	Attended CSIS 4495 class; reviewed project requirements; brainstormed initial ideas based on Richmond Hospital experience
Jan 14, 2026	1.5	Researched BC healthcare navigation tools; identified gap in existing solutions
Jan 15, 2026	2.0	Drafted initial chatbot concept; researched CTAS triage guidelines
Jan 16, 2026	3.0	Meeting #1 with instructor; received feedback to broaden scope; began expanding to multi-feature platform
Jan 17, 2026	2.5	Researched OpenAI API documentation; explored integration options
Jan 18, 2026	2.0	Meeting #2 with instructor (virtual); Option 1 approved; received guidance on focus areas
Jan 19, 2026	3.0	Designed hybrid classification approach; researched Kaggle symptom datasets
Jan 20, 2026	2.5	Created GitHub repository; set up project structure
Jan 21, 2026	3.0	Researched ML analytics techniques for dashboards; planned Power BI integration
Jan 22, 2026	2.5	Explored synthetic data generation approaches; documented data strategy
Jan 23, 2026	3.0	Drafted proposal introduction and literature review sections
Jan 24, 2026	3.5	Completed methodology section; created technical architecture diagram

Jan 25, 2026	3.0	Finalized timeline, contract, work log; reviewed complete proposal
Jan 26, 2026	2.0	Final proofreading; submitted proposal

Total Hours (Jan 13-26): 35.5 hours

This log will be updated weekly throughout the project with specific coding tasks, model training sessions, debugging, and documentation work.

7. GITHUB REPOSITORY

All project documentation and source code are maintained in a public GitHub repository for version control and collaboration.

Repository URL: <https://github.com/desporamon/BC-Health-Platform>

Folder Structure:

Folder	Contents
DocumentsAndReports/proposal/	Project proposal document
DocumentsAndReports/progress-reports/	Weekly progress reports
DocumentsAndReports/midterm/	Midterm report and demo video
DocumentsAndReports/final-report/	Final report and documentation
implementation/src/	Source code (Streamlit application)
implementation/data/	Datasets (raw and processed)
implementation/notebooks/	Jupyter notebooks for data exploration
implementation/samples/	Proof of concept code
implementation/tests/	Unit tests

Direct Link to Proposal: <https://github.com/desporamon/BC-Health-Platform/tree/main/DocumentsAndReports/proposal>

8. REFERENCES

Canadian Emergency Department Triage & Acuity Scale (CTAS) National Working Group. (2008). Implementation Guidelines for the Canadian Triage and Acuity Scale. Canadian Association of Emergency Physicians.

Ceney, A., Tolond, S., Glowinski, A., Marks, B., Swift, S., & Palser, T. (2021b). Accuracy of online symptom checkers and the potential impact on service utilisation. *PLoS ONE*, 16(7), e0254088. <https://doi.org/10.1371/journal.pone.0254088>

Holzinger, A., Biemann, C., Pattichis, C. S., & Kell, D. B. (2017). What do we need to build explainable AI systems for the medical domain? arXiv preprint arXiv:1712.09923. <https://arxiv.org/abs/1712.09923>

Hong, W. S., Haimovich, A. D., & Taylor, R. A. (2018). Predicting hospital admission at emergency department triage using machine learning. *PLOS One*, 13(7), e0201016. <https://doi.org/10.1371/journal.pone.0201016>

Lee, J., Yoon, W., Kim, S., Kim, D., Kim, S., So, C. H., & Kang, J. (2020). BioBERT: A pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4), 1234-1240. <https://doi.org/10.1093/bioinformatics/btz682>

Liao, M., Li, Y., Kianifard, F., Obi, E., & Arcona, S. (2016). Cluster analysis and its application to healthcare claims data: A study of end-stage renal disease patients who initiated hemodialysis. *BMC Nephrology*, 17, 25. <https://doi.org/10.1186/s12882-016-0238-2>

OpenAI. (2024). GPT-4 Technical Report. Retrieved from <https://openai.com/research/gpt-4>

Miotto, R., Li, L., Kidd, B. A., & Dudley, J. T. (2016). Deep patient: An unsupervised representation to predict the future of patients from the electronic health records. *Scientific Reports*, 6, 26094. <https://doi.org/10.1038/srep26094>

Semigran, H. L., Linder, J. A., Gidengil, C., & Mehrotra, A. (2015). Evaluation of symptom checkers for self diagnosis and triage: Audit study. *BMJ*, 351, h3480. <https://doi.org/10.1136/bmj.h3480>

APPENDIX: AI PROMPT HISTORY

The following documents my AI tool usage during the proposal development phase. As noted in Section 5, AI was used sparingly to clarify technical concepts I had already researched. The prompts below demonstrate that I was verifying my own understanding rather than generating ideas.

Prompt Session 1: Clarifying Hybrid Approach

Date: January 19, 2026 | Tool: Claude (Anthropic)

Context: After researching CTAS guidelines, I had the idea to use rule-based logic for critical symptoms. I wanted to verify this was a sound approach.

My Prompt:

"I'm thinking of using a hybrid approach for my symptom classification - rule-based for critical symptoms like chest pain, and ML for less obvious cases. Is this a valid approach? I got this idea from reading about how CTAS works."

Summary of Response:

AI confirmed this is a valid approach used in real clinical decision support systems. It mentioned the benefit of safety (critical symptoms don't rely on ML) and flexibility (ML handles ambiguous cases).

Value I Added:

I had already decided on this approach based on my CTAS research. The AI response validated my thinking. I then independently designed the specific rules based on CTAS Level 1-2 criteria.

Prompt Session 2: Understanding OpenAI API Options

Date: January 17, 2026 | Tool: ChatGPT

Context: I knew I wanted to use OpenAI for NLP but wasn't sure about the technical implementation options.

My Prompt:

"For my healthcare chatbot, I want to use OpenAI to understand user symptoms. Can you explain how the API integration would work? I've read the OpenAI documentation but want to make sure I understand it correctly."

Summary of Response:

AI explained that OpenAI can extract structured data from unstructured text, handling variations in how users describe symptoms. It showed a basic example of API call structure.

Value I Added:

I used this understanding to design the integration flow: OpenAI extracts symptoms → my rule-based layer checks for critical patterns → my ML model classifies urgency. All code logic and prompt engineering will be my own work during implementation.

Prompt Session 3: ML Analytics Techniques

Date: January 21, 2026 | Tool: Claude (Anthropic)

Context: Based on my Power BI experience at VCH, I knew I wanted to add analytics. I wanted to understand which ML techniques would be appropriate for health data.

My Prompt:

"For my health dashboard, I want to show insights beyond just basic counts. What ML techniques could I use to analyze symptom patterns across users? I'm familiar with clustering from my ML course."

Summary of Response:

AI mentioned several techniques: K-means clustering for user grouping, anomaly detection for unusual spikes, time series for trends, and association rules for symptom co-occurrence.

Value I Added:

I selected specific techniques based on my existing knowledge: Association rules (learned in CSIS 3275), clustering (learned in ML course), and time series (comfortable from VCH work). I decided which techniques to apply to Feature 2 vs Feature 5 based on the individual vs population scope.

Prompt Session 4: Data Source Question

Date: January 19, 2026 | Tool: Claude (Anthropic)

Context: My instructor asked about data sources for training the ML model. I wanted to verify my plan to use public datasets.

My Prompt:

"My instructor asked where I would get training data to classify symptom severity. I'm planning to use Kaggle datasets. Is this a reasonable approach for an academic project?"

Summary of Response:

AI confirmed Kaggle symptom-disease datasets are commonly used for academic ML projects. It noted I would need to manually map diseases to urgency levels since datasets typically map symptoms to diseases, not directly to triage urgency.

Value I Added:

I independently researched and identified specific Kaggle datasets. I will create my own urgency mapping based on CTAS guidelines (e.g., heart attack symptoms → Emergency, common cold → Self-care).

Prompt Session 5: Synthetic Data for Demo

Date: January 22, 2026 | Tool: Claude (Anthropic)

Context: I realized my dashboards would be empty at launch with no user data. I asked about standard practices.

My Prompt:

"For my demo, the dashboards will be empty since there's no user data yet. What do developers typically do in this situation for proof-of-concept applications?"

Summary of Response:

AI explained that synthetic/seed data is standard practice for demos. It mentioned generating realistic fake records that enable ML analytics to show meaningful patterns.

Value I Added:

I decided to generate 5,000 synthetic records. I will design the data generation script myself, including realistic symptom patterns and an embedded 'flu outbreak' trend so the ML analytics have patterns to detect.

Prompt Session 6: Document Formatting

Date: January 25, 2026 | Tool: Claude (Anthropic)

Context: I needed help formatting my proposal document with proper tables and structure.

My Prompt:

"Can you help me format my proposal into a Word document with proper tables? I have all the content written, I just need help with the formatting."

Summary of Response:

AI helped structure the document with appropriate headings, tables, and formatting consistent with academic standards.

Value I Added:

All content, ideas, methodology, and technical decisions are my own. AI was used purely for document formatting assistance - similar to using Microsoft Word's formatting features or asking a classmate for formatting help.

Prompt Session 7: System Flow Diagram Generation

Date: January 25, 2026 | Tool: Claude (Anthropic)

Context: I had already designed the 3-layer hybrid classification system (OpenAI → Rules → ML) and written the technical explanation for Section 2.3.2. I needed a professional-looking diagram to visualize the flow I had designed.

My Prompt:

"Create a professional healthcare system flow diagram for a '3-Layer Hybrid Classification System.' The flow should show: User Input at top, then Step 1 (OpenAI API for NLP understanding), then Step 2 (Rule-Based Safety Check using CTAS), then a decision diamond for 'Critical Match?', branching to Emergency (Call 911) if YES, or Step 3 (ML Model - Random Forest classification) if NO, and finally a Final Output box showing urgency level, explanation, and BC Healthcare Resources. Use a clean teal color scheme with rounded rectangles."

Summary of Response:

AI generated a visual flowchart based on my specifications, rendering the system flow I had designed into a professional-looking diagram.

Value I Added:

I designed the entire 3-layer hybrid architecture myself based on my CTAS research and instructor guidance. The flow logic, component names, decision points, and technical details (OpenAI, CTAS, Random Forest, urgency levels) were all my own design. AI was used solely as a visual rendering tool—similar to using Draw.io or PowerPoint to create a diagram from my specifications.

Prompt Session 8: Architecture Overview Diagram Generation

Date: January 25, 2026 | Tool: Claude (Anthropic)

My Prompt:

"Create a professional technology architecture diagram for a healthcare web application called 'BC Personal Health Management Platform.' Show: User (BC Resident) connecting via Internet to Streamlit Cloud application containing 5 features (Symptom Chatbot, Personal Dashboard, Facility Finder, Education Library, System Analytics). Include Core Processing Components (Python 3.10+, NLTK/spaCy, scikit-learn ML Engine, Rule-Based Logic). Show external services: OpenAI API and Power BI Service. Show data layer: SQLite, CSV Export, GitHub. Show data sources: Kaggle, HealthLinkBC, CTAS. Use teal color scheme matching enterprise software documentation."

Summary of Response:

AI generated a visual architecture diagram based on my detailed specifications, rendering all the components and connections I had specified.

Value I Added:

I designed the complete system architecture myself, determining which technologies to use for each component based on my coursework and VCH co-op experience. The technology choices (Streamlit, OpenAI, scikit-learn, Power BI, SQLite), data flow design, and integration approach were all my own decisions developed over weeks of planning. AI was used only to render my architecture design into a professional visual format—equivalent to using a diagramming tool like Visio or Lucidchart.

Summary of AI Usage

Total AI Sessions: 8 sessions over 9 days (Jan 17-26)

Primary Purpose: Clarifying and validating concepts I had already researched and creating flow diagrams

NOT Used For: Generating project ideas, writing content, making design decisions

Key Point: In each prompt, I am basically asking questions about MY ideas or confirming MY understanding. I got the idea for the project from my experience at Richmond Hospital. The way to do it technically came from my coursework (CSIS 3275, CSIS 3400, ML course) and VCH co-op experience. AI was just like a buddy to hash out ideas, kind of like talking to a classmate or going to office hours for help.